

Technical Specification & Engineering Cut Sheet

Equipment Cut Sheet for Environmental Product Declaration (ISO 14025 / EN 15804+A2)

Product Name: Carrier AquaEdge 19DV Centrifugal Chiller
Model Capacity: 500 RT (1,758 kW Nominal Chilled Water)
Refrigerant Type: R134a (Initial Factory Charge: 45.0 kg, fugacity leak rate: 2.0%/yr)
Plant Utility Power: 34,000 kWh annual manufacturing electricity allocated to unit
PCR Category Standard: UL 10010-4 Part B & EN 15804+A2 (Industrial HVAC Systems)
Reference Service Life (RSL): 25 Years with standard maintenance intervals

Component	Material	Weight (kg)	Supplier	Transport (km)
Compressor Volute & Casting	Structural Steel	2,250	Midwest Iron Works	420
Evaporator & Condenser Tubes	Copper	720	Great Lakes Copper Corp	280
Semi-Hermetic Induction Motor	Electric Motor	480	ElectroMotors Industrial	650
Shell Thermal PUF Insulation	Polyurethane Foam	140	FoamTech Insulation	190
Power Inverter & Solid-State VFD	Power Electronics	110	Advantech Power Systems	890
Microchannel Subcooler Coil	Aluminium Alloy	95	Alloy Heat Exchangers	350
Refrigerant High-Pressure Piping	Stainless Steel 304	85	Precision Alloy Tubing	340
Base Skid Frame & Mounts	Structural Steel	540	ArcelorMittal Midwest	260